



E/I/I test tables

Haul truck and light vehicle collision, Bindarra Gold Mine

Exported 30 September 2026

INVESTIGATION WORKFLOW SUITE

E/I/I test tables: Haul truck and light vehicle collision, Bindarra Gold Mine

Existence: did it exist or occur? Influence: did it influence this occurrence? Importance: is it worth addressing regardless?

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	MSG-14 clause 3.2 requires the traffic management plan to be reviewed within twelve months and after any significant change to the road network, and defines neither the term nor an example of one. Clause 5.1 requires the operator to nominate the intersections and interaction zones at which positive communication is required, and sets no minimum and no basis for the nomination. Clause 5.4 requires a light vehicle to be detectable by a heavy vehicle operator and offers five alternative means, any one of which satisfies it. Clause 3.5 provides that the plan is a site document, not approved by and not submitted to the Inspectorate except on request after an occurrence. Bindarra's plan met the guideline on its face on the day of the occurrence: it had been reviewed inside twelve months, it nominated nine interaction zones, and it provided for detectability by line of sight and by elevated marking. The mine was not in breach.
Influence	Pass - likely (70%)	Three separate terms in the one guideline are left to the site to fill in, and at Bindarra all three were filled in by the same document, reviewed once a year, with no outside eye on any of it.
Importance	Not tested	(none)

Investigator notes: The mine is not in breach of anything, which is exactly why this belongs in External Influences rather than in a compliance finding. Be careful in the report not to let this read as blaming the regulator; the point is the shape of the discretion, not a failure by the Inspectorate.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	CRL-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant is introduced. WO-26-4471 cut 240 m of access to bore DW-14 at RL 312, classified the work as routine earthworks within the approved Stage 4 design, recorded management of change as not required under clause 5.3, and left its handover field reading N/A. The three referrals that a management of change record would have triggered at clause 5.4 are the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system geofences. None of the three was made. The in-cab road network published on 24/07/2026 was still current on the night and shows the main ramp with no intersection at RL 312. Of the fourteen new

		roads cut in Stage 4 since the last plan review, the three raised under a management of change record were referred to the plan's owner and none of the eleven raised under a works order was.
Influence	Pass - very likely (90%)	A works order built an intersection into a haul road and the classification on its front page removed every consequence of having done so. Eleven roads have been built the same way since the last review.
Importance	Not tested	(none)

Investigator notes: The clearest single organisational finding in the investigation. Everything that should have followed a new road into an operating pit lives inside the management of change record, and the exclusion at clause 5.3 removed the record.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	CRL-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement, and Merrivale Survey & Geotechnical has held one since 2023. Clause 7.4 states that equivalence does not extend to the Bindarra site-specific supplement or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive in an operating pit and shall not be substituted. Clause 7.5 makes the engaging department responsible for verifying 7.4 before a card is issued. The gatehouse issues cards against the recognition agreement alone, holds no record of either item, and is not asked for one; nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not. The driver's training file records the Merrivale online module of 24/07/2026 and a 20 minute gatehouse briefing, and no supplement and no pit driving assessment. He had never seen BIN-TMP-004 and did not know it existed.
Influence	Pass - likely (80%)	A person was driving a light vehicle on an active haul road on his third night at the mine, without the assessment the site's own standard says may not be substituted, and no part of the system that let him through was looking for it.
Importance	Not tested	(none)

Investigator notes: Watch the wording of this in the report. The standard is right and says the right thing; what failed is that the only place the verification could have happened has no field for it. Do not write it as a failure by the driver.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The collision awareness system on the Stage 4 haul fleet operates at Level 7, operator advisory; Level 9 intervention is available on the fleet and is not enabled at Bindarra. Advisory alerts are raised on a tag entering the 60 m ring above a 10 km/h closing rate, except inside the S4-LOAD suppression geofence, which was drawn around the loading face on 14/03/2025 to stop nuisance alerts and extended northward on 09/06/2026 when the face advanced. The extended boundary lies 180 m up-ramp of the RL 312

		intersection, placing the intersection inside the suppressed area. The extension was made as a system configuration task by the contractor's technical services group, is not recorded in the traffic management plan, and was not reviewed against it. The log records a proximity event at 18:47:09.2 at 21.4 m and 13 km/h closing, alert suppressed. In the twelve months before the occurrence HT-407 raised 412 advisory alerts, every one outside the geofence and none inside it, while 68 proximity events inside it were suppressed, nine of them a light vehicle tag on the main ramp below the boundary. No training record anywhere on site, for any person, covers the geofences.
Influence	Pass - likely (75%)	Nine earlier events of exactly this kind were recorded by the system, visible to anyone who queried it, and reported to nobody, because the standing report covers alert counts and system health and does not report suppressions at all.
Importance	Not tested	(none)

Investigator notes: Both the operator and the supervisor raised this unprompted and the statistics bear them out. The finding is not that the geofence exists; nuisance alerts around an excavator will destroy a system. The finding is that its boundary moved 180 m in June and nobody outside the group that moved it was told.

One emergency response paramedic rostered on night shift, with the second responder called from the village

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The Bindarra night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication is begun, is called in from the accommodation village 11 km from the gatehouse. On the night, dispatch paged the team at 02:48, the first response vehicle entered the pit at 02:51 and reached the scene at 02:54, the second responder arrived with the hydraulic rescue tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The post-incident review found the extrication delayed by approximately 9 minutes waiting for the second responder and the tool, and recorded that the delay did not affect the clinical outcome because the injuries were stable and the patient was conscious throughout.
Influence	Fail - unlikely (10%)	A nine minute wait for the second half of an extrication capability is a defence that is not there when it is needed, and nothing about the arrangement changes with the severity of the casualty it is called to.
Importance	Pass - very likely (90%)	It did not influence this occurrence and it is still a finding. The driver was conscious, stable, restrained and protected by an intact roll-over structure, and the delay cost him nothing. Change one thing, a fuel fire, an unconscious casualty, an airway that needed the vehicle opened in the first ten minutes, and the same nine minutes is a different outcome. The gap that matters is structural rather than clinical: the site's extrication capability on night shift is a single person plus a drive from the village, and no part of the arrangement scales with what has happened.

Investigator notes: Left in draft deliberately. The question of what the roster looks like across the other three night shifts on site is out with the emergency response coordinator and this object should not be finalised until that answer is in, even though its own tests are settled.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Test	Result	Supporting text
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Existence	Pass - virtually certain (100%)	BIN-TMP-004 clause 4.3.1 requires a positive call on the pit channel before entering any intersection listed at Appendix B, and 4.3.2 requires no call anywhere else. Appendix B lists nine intersections. Its own note provides that the appendix is reviewed and reissued at the annual review, and that an intersection is added where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication. Schedule 1 is dated 12/11/2025 and is reissued at the annual review. No provision anywhere in the plan allows an intersection to be added between reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. The RL 312 spur is on neither document. Eleven of the fourteen new Stage 4 roads since the last review were never referred, and no out-of-cycle review has been held in the plan's eleven year history. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently: calls are made at the nine intersections on the crib room board and nowhere else.
Influence	Pass - very likely (90%)	This is the mine's own defence against exactly this interaction, and at this intersection it was unarmed by its own terms. The next date on which the spur could have been listed is 12/11/2026, three months after the collision.
Importance	Not tested	(none)

Investigator notes: The safety issue the map exists to find. The control is sound and its trigger is circular: an intersection reaches the list through a drawing that is only reissued at the review, and nothing requires a new road to be referred at all. Recommend this one in its own right whatever else the report says.

Local Conditions

Dust on the ramp reduced the truck operator's ability to see the light vehicle

Test	Result	Supporting text
Existence	Inconclusive - about as likely as not (45%)	For: the water cart's last pass on that section was 1 hour 37 minutes before the occurrence against a standing 90 minute interval, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the ten minutes covering the occurrence against a shift mean of 88. The truck operator described dust hanging on the ramp, said the cart had been through earlier and it dries out, and the descent was behind his own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Left in draft on purpose. Two honest accounts that do not agree, one measurement that shows a deviation and not an effect, and no way to close it with what exists. Say so in the report rather than picking a side.

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The operator was on night 10 of a 14 night swing, working 18:00 to 06:30 local, and the collision was at 02:47, 10 hours 47 minutes into the shift and inside the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met: no more than 14 consecutive nights, a minimum 10 hour break, and no shift extended beyond 12 hours 40 minutes in the preceding nine. His own voluntary sleep record gives 6 hours 30

		minutes in the day before and between 5 and 7 hours on each of the preceding nine days. He described himself in interview as tired in the way you are on night ten, not falling asleep, awake and driving properly.
Influence	Fail - unlikely (25%)	Ten consecutive nights at the bottom of the circadian cycle is a standing exposure built into the roster rather than a one-off, and it is worked by every operator on this pit on every swing.
Importance	Pass - likely (85%)	It failed Influence for this occurrence and remains a finding worth acting on. The Stage 4 night roster puts every haul truck operator at the wheel of a 220 tonne vehicle in the small hours of a tenth consecutive night, at a mine with no fixed lighting anywhere in the pit, where the sole defence against an unlit hazard at an unlisted intersection is the operator's own eyes. Nothing about that changes because it did not bite here, and the same roster is worked on three other pits on this site.

Investigator notes: The tempting object. It exists, it reads as a cause, and the evidence will not carry it for this occurrence. Recording it as an other risk factor is the honest answer and it is still going in the report.

The ramp surface was wet and the truck could not stop in the distance available

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because an 18 m push down a 9 per cent grade invites a question about the surface. It fails. The met station records no precipitation on the night and the last rainfall at the site was 26 days before the occurrence. The water cart log puts the last pass on that section at 17:10 UTC, 1 hour 37 minutes before, and the standing night shift instruction is a pass every 90 minutes, so the surface had been drying for longer than the interval rather than freshly watered. The investigator examined the ramp at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The braked tyre scuff runs continuously for 62 m with no sign of a locked or sliding wheel, and the download gives a mean deceleration of 1.85 m/s ² against the unit's own tested 1.9 m/s ² on a dry 9 per cent grade. There is no evidence of a wet or slippery surface from any source.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: Raised early because a 220 tonne truck sliding down a wet ramp is the first thing anybody pictures. Disposed of in a morning and kept on the map so the report can say it was tested rather than assumed away.

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The survey pickup of 13/08/2026 places the spur mouth 34 m beyond the high point of a crest in the ramp. An operator descending at an eye height of 5.80 m first has line of sight to the spur mouth at 41 m; a driver at the spur mouth at an eye height of 1.45 m first has line of sight to a descending truck at 38 m, with the 1.6 m inner windrow taking the truck's lower body out of view for the last 25 m of the spur's approach. BIN-TMP-004 clause 6.2 requires available sight distance at every intersection to be not less than the stopping distance of the largest vehicle at the posted speed and in no case less than 120 m, and requires a stop, signalised or controlled intersection where that

		cannot be achieved. The stopping distance of a 220 t class truck empty at the posted 40 km/h on a 9 per cent descending grade, on the manufacturer's data with a 1.5 second response, is 96 m. The intersection meets neither limb of clause 6.2. It carried no warning sign, no give-way control, no delineation and no posted speed, because clause 6.4 applies only to intersections shown on the road layout drawing and this one is not on it. Seven days earlier a dewatering technician reported the same sight line from the same position in his own words.
Influence	Pass - very likely (90%)	41 m against a 96 m stopping distance means that a truck at the posted speed cannot stop for anything at this intersection, whatever the operator does and however alert he is.
Importance	Not tested	(none)

Investigator notes: The measurement the organisational half of the map rests on. Make sure the report draws the line between this number and the stopping distance: at 41 m, attention, training and a radio call are all downstream of a geometry that cannot be driven safely.

Individual Actions

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The driver's account, corrected by him on read-back from stopped to slowed, is that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out; he does not believe he came to a complete stop. The scene evidence is consistent: the impact is 6.2 m into the ramp from the eastern edge of the spur mouth, and the gouge runs from there. BIN-TMP-004 clause 4.1 requires light vehicles to give way to heavy vehicles at all times, which is the rule he described himself as applying. There was no give-way control, no stop line, no warning sign and no posted speed at the mouth, because clause 6.4 applies only to intersections on the road layout drawing. From the mouth, at his eye height, the survey gives 38 m of sight line, and the windrow takes the truck's lower body out of view for the last 25 m of the approach, so a look left from the stopped position would have shown him approximately what a look at 25 km/h showed him. He had done no Bindarra pit driving assessment and had never seen the traffic management plan.
Influence	Pass - very likely (90%)	This is the action that put a light vehicle across a descending haul truck's path, and it is the one place on the map where a different decision in the last ten seconds would have avoided the collision.
Importance	Not tested	(none)

Investigator notes: The object the report must handle most carefully. He is the injured party, he corrected himself on the point least helpful to him, and there was no sign, no line and no assessment to tell him otherwise. Write it as an action, not a failing.

Neither operator made a positive call on the pit channel before entering the intersection

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The pit channel was logged continuously and the investigator listened to the whole period from the shift's return from crib to two minutes after the collision. There is no transmission by either vehicle about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. LV-118

		made no transmission on the channel at all that shift beyond logging on at 17:04 UTC. No call was required: BIN-TMP-004 clause 4.3.2 requires none at an intersection not listed at Appendix B, and this one is not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently and without being led: calls are made at the nine intersections on the crib room board and at no others, and the supervisor said plainly that he has never corrected anybody for not calling elsewhere and that requiring calls everywhere would make the channel unusable.
Influence	Pass - likely (85%)	The one action that would have put each vehicle in the other's knowledge was available, cost nothing, and was on neither operator's mind, because no document put it there and the local practice matched the document exactly.
Importance	Not tested	(none)

Investigator notes: Careful with the wording of this one. It is an omission with no owner: the plan expressly requires no call there, every operator on the pit described the same practice, and the supervisor confirmed he has never pulled anybody up for it.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	The download records the last steady data point at 18:47:06 at 37 km/h, 41 m above the intersection, and service brake pressure rising from 0 to 620 kPa in 0.4 seconds from 18:47:07.4, with the retarder taken to maximum and 14 degrees of right steering input in the same movement. That is a 1.4 second response, inside the 1.5 seconds the plan's own stopping distance calculation allows. Mean deceleration from the brake application to the collision was 1.85 m/s ² against the unit's tested 1.9. Impact speed was 24 km/h; with no braking it would have been 37 km/h. The scene shows the braked scuff deviating 1.4 m right over its first 20 m, into the wall side, which is the only direction that did not take the light vehicle down a 100 m drop with him. He stopped 19 m below the intersection, called the collision, followed the dispatcher's instruction to stay in the cab until the ramp was stopped, then went down and stayed with the driver until the response team arrived.
Influence	Pass - very likely (90%)	The difference between a 37 km/h and a 24 km/h impact on the side of a light vehicle is the difference between the injuries actually sustained and a very different outcome. The steering decision is what kept the vehicle against the inner windrow rather than over the edge.
Importance	Not tested	(none)

Investigator notes: Recorded as a positive event because it is one. The same investigation that finds a 41 m sight line should say that what the operator did inside those four seconds is the reason this is a survivable injury and not a fatality.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

Test	Result	Supporting text
Existence	Pass - extremely likely (95%)	HT-407 was descending the ramp empty at 37 km/h with no intersection shown on its in-cab road network, no warning sign on the approach, no call on the channel and no alert from a collision awareness system that had logged the light vehicle at 21.4 m. LV-118 was leaving the spur at about 25 km/h with 38 m of sight line, no give-way control, no call and no knowledge of the truck. The first sighting available to either of them was at 41 m and

		about four seconds, which is what the survey predicts for the geometry. Neither the radio log, the download, the CAS log nor any of the three interviews contains a single indication that either operator knew the other was there before that moment. The night shift supervisor did not know a light vehicle was in the pit at all, and there is no permit, call-in or record by which he could have.
Influence	Pass - extremely likely (95%)	Every organisational and control finding on this map converges on this one condition: two vehicles in the same place, at night, each invisible to the other until four seconds remained.
Importance	Not tested	(none)

Investigator notes: The convergence point. Every chain on this map arrives here, which is why the report's analysis should treat it as one mechanism rather than five separate findings.

Light vehicle driver seriously injured in a collision with a haul truck on the main ramp

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	At 18:47:12 UTC on 11/08/2026, 02:47 local, a 220 t class rigid haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine empty at 24 km/h struck the right side of a light vehicle emerging from the RL 312 dewatering spur, pushed it 18.4 m down the ramp and rolled it onto its left side against the inner windrow. The light vehicle carried one occupant, a grade control technician employed by a survey subcontractor, on his third night shift at the mine. He was conscious throughout, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre at 07:40 local.
Influence	Pass - virtually certain (100%)	A serious injury: a fractured left clavicle, three fractured left ribs and a small pneumothorax, treated at the surface medical facility and evacuated by air. The light vehicle was destroyed; its roll-over protection structure was deformed inward at the right rear mounting and the seatbelt webbing carried loading marks, so two defences were loaded to the point of showing it. The haul truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.
Importance	Not tested	(none)

Investigator notes: The critical event. Everything else on this map exists to explain it.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	Impact at 18:47:12.0 at 24 km/h, 4.6 seconds after the brake application and 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 is concentrated on the right side from the B pillar forward; damage to HT-407 is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right side glass and mirror fragments lie from 4.8 m to 14 m down-ramp of the impact position.
Influence	Pass - virtually certain (100%)	A 220 tonne vehicle striking the side of a 3 tonne one leaves the outcome entirely to the light vehicle's structure, the restraint and the speed, and two of those three were already settled before the sighting.

Importance	Not tested	(none)
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Investigator notes: The link to make sure the report draws: the truck struck the side of the vehicle, not the front, which is what the intersection geometry makes inevitable and what the roll-over protection then had to deal with.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

Test	Result	Supporting text
Existence	Fail - very unlikely (5%)	Hypothesis tested because a 220 tonne truck that does not stop in the distance available raises the brakes first. It fails. The monthly brake performance test on 03/08/2026, eight days before, gave a mean deceleration of 1.9 m/s ² in the empty condition on a 9 per cent grade. The test repeated on 14/08/2026 before the unit returned to service gave 1.92. The download of the occurrence itself gives 1.85 over the 4.6 seconds of braking, which sits between the two tests and inside normal scatter. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder was at maximum throughout, and no brake, retarder or park brake defect was raised against the unit in the twelve months before the occurrence. The pre-start check sheet for 11/08/2026 is complete and records no defects. The braking system performed to specification before, during and after the occurrence.
Influence	Not tested	(none)
Importance	Not tested	(none)

Investigator notes: The first hypothesis anyone offers when a haul truck does not stop, and the first one the records kill. Kept on the map because the report should show it was tested, not assumed away.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

Test	Result	Supporting text
Existence	Pass - virtually certain (100%)	A continuous gouge runs 18.4 m down-ramp from the impact position, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp, 18.4 m below the impact. The roll-over protection structure is deformed inward at the right rear mounting and was not breached. The driver's seatbelt was found fastened with loading marks on the webbing. The occupant sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The door on the resting side was against the ground, so extrication was through the rear of the vehicle.
Influence	Pass - virtually certain (100%)	The injuries are the ones a side impact and a roll-over produce when the structure holds and the restraint is worn. The same sequence with either of those absent is a different report.
Importance	Not tested	(none)

Investigator notes: The mechanism of injury, and the place to say what worked: the restraint was fastened and loaded, and the roll-over structure deformed without being breached. Both belong in the report.